

Bari Category Excellence Intelligence Framework: A Multi-Dimensional Analytical System for Food Quality and Authenticity

The Metropolitan City of Bari, in collaboration with international institutions such as CIHEAM Bari, has pioneered the operationalization of sustainable food systems and quality authentication schemes in Southern Italy. Under regional frameworks like the "Prodotti di Qualità di Puglia" voluntary certification and the academic guidelines of "Med Diet 4.0," agricultural policies are shifting from a purely production-centric focus to an integrated consumer-health and authenticity-verification standpoint.

To support this transition, the Bari Category Excellence Intelligence Framework provides an automated, scalable methodology to rate and verify the quality and authenticity of major food categories. This intelligence engine utilizes a synthesis of the Nutrient Quality Score (NQS 11.1) and the Mediterranean Index (MI) to generate actionable quality scores using consumer-facing label data, while implementing digital mitigation proxies for chemical and structural attributes that remain invisible without laboratory intervention.

Architectural Overview of the Bari Scoring Engine

The Bari scoring engine evaluates food products through a standardized, multi-dimensional algorithm that weights raw material purity, processing indicators, and regional certifications. The core score calculation balances nutritional density against industrial processing indicators:

$$\text{Bari Category Excellence Score} = \alpha \left(\text{Nutritional Integrity} \right) + \beta \left(\text{Processing Fidelity} \right) + \gamma \left(\text{Authenticity Verification} \right)$$

Where the parameters represent:

- **Nutritional Integrity:** Derived from portion-based daily reference values, fiber-to-carbohydrate ratios, and the absence of refined sugars.
- **Processing Fidelity:** Penalizes ultra-processed markers, such as artificial emulsifiers, synthetic preservatives, and high-heat case hardening.
- **Authenticity Verification:** Rewards Protected Designation of Origin (PDO/DOP), Protected Geographical Indication (PGI/IGP), and state-level organic certifications.

Category-Specific Quality and Authenticity Intelligence

1. Bread (Sourdough Focus)

The physical chemistry of sourdough is governed by a symbiotic co-fermentation of wild yeasts (e.g., *Saccharomyces exiguus*) and heterofermentative lactic acid bacteria (LAB) (e.g., *Lactobacillus sanfranciscensis*). During this biological window, LAB metabolize simple starches to generate lactic and acetic acids, lowering the pH to a range of 3.8 to 4.5. This acidic environment activates endogenous cereal proteases, which break down gluten proteins into easily digestible peptides and reduce the bread's overall glycemic index.

Exceptional sourdough relies on stone-ground heirloom grains (such as Spelt, Senatore Cappelli, or Einkorn) and a cold-proofing phase lasting 24 to 48 hours to optimize gluten degradation and prebiotic synthesis. Average industrial bread substitutes this biological duration with dehydrated sourdough powders or hybrid bakers' yeast to accelerate the rise to under 4 hours, preserving the native, highly crystalline starch structures that lead to glycemic spikes. Poor bread—termed "fake sourdough"—bypasses biological fermentation entirely, relying on chemical acidulants like citric acid, vinegar, and synthetic preservatives such as calcium propionate to mimic the organoleptic profile of authentic fermentation.

Bari can capture ingredient decks to penalize non-biological additives, but cannot directly measure the actual yeast-to-LAB microflora ratio or the resulting organic acid profile. To mitigate this, the engine flags products containing yeast or citric acid, while rewarding clean-label designations of flour, water, salt, and starter.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Stone-ground heirloom or ancient grains (e.g., Einkorn, Spelt); organic flours.	Standard conventional refined wheat flour; added commercial vital gluten.	Low-grade, highly processed white flour with flour-treatment agents.
2. Production & Processing	24–48 hour cold fermentation with active, live wild starter cultures.	3–4 hour bulk rise using commercial yeast and dehydrated sourdough starter.	Accelerated mechanical baking with zero biological fermentation time.
3. Authenticity Indicators	100% natural wild starter claims; zero	Sourdough flavor claims paired with	Chemical additives mimicking

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
	added yeast or commercial raising agents.	commercial yeast in the ingredient deck.	sourdough acidity without starter culture.
4. Fraud Risks	Misrepresenting fast-risen industrial yeast bread as traditional wild sourdough.	Minor addition of active starter to standard yeast dough to validate labeling.	Use of vinegar, citric acid, or inactive sourdough powders to fake tanginess.
5. Chemical / Measurable	Balanced lactic-to-acetic acid ratio; low glycemic index; high gluten hydrolysis.	Moderate organic acid profile; high glycemic index; intact, non-hydrolyzed gluten.	High glycemic index; detectable citric/acetic acid; presence of calcium propionate.
6. Sensory Quality	Deeply blistered, caramelized crust; highly irregular crumb; complex yogurt-like acidity.	Uniform, spongy crumb; faint yeasty aroma with flat, short-lived acidity.	Spongy, soft texture; sharp, synthetic vinegar-like odor; resists natural molding.
7. Label-Visible Signals	Simple deck: only flour, water, salt, and wild starter; certified organic logos.	"Sourdough" name with "yeast," "malt extract," or "ascorbic acid" on the deck.	"Citric acid," "acetic acid," "calcium propionate," or "monoglycerides" listed.
8. Label-Invisible Signals	Active yeast-to-LAB cellular ratio; native enzyme activity; absolute fermentation time.	Starter culture purity; presence of trace chemical bleaching agents.	Total absence of lactic acid bacteria DNA or native fermentation byproducts.
9. Bari Scoring Signals	Score: +3 for flour/water/salt/starter	Score: 0 for baseline standard	Score: -5 for presence of citric/acetic acid or

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
	only; +2 for verified organic certification.	yeast-leavened flour products.	chemical preservatives.

2. Butter

Butter quality is fundamentally determined by the pasture-based diet of the dairy herd and the mechanical preservation of the emulsion during churning. Exceptional butter is derived from 100% grass-fed, pasture-raised cows, which naturally yields elevated concentrations of beta-carotene, Omega-3 fatty acids, and conjugated linoleic acid (CLA). It is slowly fermented with lactic cultures to generate high concentrations of diacetyl, providing a rich, nutty, and sweet cream aroma.

Average butter is uncultured sweet cream butter sourced from conventional, grain-fed herds. It lacks diacetyl development and displays a pale yellow or white hue, often standardized artificially with natural colorants like annatto. Poor butter contains added water-binding emulsifiers, vegetable oils, or palm-derived fats to lower raw material costs. It exhibits a greasy, structurally unstable emulsion that separates into water and oil phases rapidly when heated.

Bari can capture pasture-fed, organic, and PDO/PGI designations while immediately penalizing any ingredient list containing additives beyond "cream" and "salt". It cannot directly verify the CLA fatty acid profile or detect low-level palm oil blending, which requires gas chromatography, but the scoring engine applies a risk flag to conventional, non-organic butter lacking grass-fed certifications.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Fresh cream from 100% grass-fed, pasture-raised cows.	Pasteurized sweet cream from conventional, grain-fed dairy herds.	Low-grade dairy fat blended with reconstituted milk solids or palm oil.
2. Production & Processing	Slow biological ripening with active lactic starters; traditional batch churning.	Direct pasteurization and rapid continuous mechanical churning; uncultured.	High-speed industrial processing with added emulsifiers and water injection.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
3. Authenticity Indicators	Certified grass-fed claims; organic or regional PDO/PGI certifications.	Standard dairy origin labels without pasture or cultivation declarations.	Margarine-like yellow spreads containing modified dairy ingredients.
4. Fraud Risks	Blending with cheap vegetable fats or palm-based triacylglycerols.	Adding synthetic diacetyl to non-ripened butter to mimic culturing.	Watering down fat content using water-binding starches and gel stabilizers.
5. Chemical / Measurable	High CLA and Omega-3 content; natural beta-carotene levels; zero additives.	Standard milk fat profile; moderate oxidation; zero active bacterial enzymes.	Low milk fat percentage; detectable palm fat markers; presence of emulsifiers.
6. Sensory Quality	Rich, nutty aroma; deep natural gold color; smooth, spreadable, wax-like texture.	Mild, flat milky flavor; pale yellow hue; firm, brittle cold texture.	Greasy mouthfeel; off-flavors (rancid/cheesy); oil-bleeding when stored at 20 [°] C.
7. Label-Visible Signals	Ingredient deck: cream (or milk) and salt only; "100% Grass-Fed" logos.	Sweet cream and salt; lack of organic or grass-fed pasture claims.	Added "mono- and diglycerides," "annatto," "water," or vegetable oils.
8. Label-Invisible Signals	True ratio of Omega-3 to Omega-6 fatty acids; absence of pesticide residues.	Exact grass-fed percentage of the herd's diet.	Trace concentrations of palm kernel oil or hydrogenated seed fats.
9. Bari Scoring Signals	Score: +3 for "100% Grass-Fed" or pasture	Score: 0 for basic clean-label sweet cream butter.	Score: -5 for presence of emulsifiers,

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
	claims; +2 for PDO/DOP seals.		stabilizers, or non-dairy lipids.

3. Olive Oil

Extra Virgin Olive Oil (EVOO) represents the nutritional core of the Mediterranean Diet. Exceptional EVOO is first cold-pressed under 27°C (85°F) within 2 hours of harvest to minimize cellular oxidation and lipolytic breakdown. This processing speed yields a free fatty acidity (FFA) of $\leq 0.3\%$ (well below the regulatory cap of 0.8%) and a peroxide value (PV) of $\leq 10 \text{ meq O}_2/\text{kg}$ (average 4.9). Advanced quality indicators confirm freshness, with diacylglycerols (DAGs) $\geq 90\%$ and pyropheophytins (PPP) $\leq 5\%$ proving the absence of old or heat-treated oils. High polyphenol content ($>250 \text{ mg/kg}$) provides sensory bitterness and a sharp, peppery throat pungency.

Average EVOO meets basic International Olive Council (IOC) standards (FFA $\leq 0.8\%$, PV $\leq 20 \text{ meq O}_2/\text{kg}$) but lacks high-phenolic potency, exhibiting mild fruitiness with zero bitterness or pungency. Poor olive oil consists of oxidized or deodorized lampante oils blended with refined olive, hazelnut, or cheaper seed oils. These oils show elevated ultraviolet absorption ($K_{270} > 0.22$) and the presence of secondary oxidation compounds.

Bari can capture protective dark glass packaging, specific harvest dates (verifying freshness within 12–18 months), and certification seals such as COOC, USDA, or PDO/PGI. It cannot measure FFA, PV, DAGs, or PPP values directly, which requires UV spectrophotometry and chromatography. The scoring engine mitigates this by heavily penalizing clear glass packaging and oil stored past its optimal freshness window.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Olives cold-pressed within 2 hours of harvest; pristine, unbruised fruit.	Olives milled within 24–48 hours of harvest; standard	Overripe, bruised, or insect-damaged olives stored

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
		commercial fruit.	in warm heaps.
2. Production & Processing	Cold extraction below 27 ^{°C} (85 ^{°F}); nitrogen blanketing in storage.	Standard commercial cold pressing with minimal oxidative protection.	Prolonged milling times; high processing temperatures; thermal deodorization.
3. Authenticity Indicators	COOC, USDA Organic, or regional PDO/PGI certifications with serial tracking.	Basic Extra Virgin label without regional or premium certification seals.	Blended oils containing refined pomace, hazelnut, or seed oils.
4. Fraud Risks	Adulteration with soft-column refined oils or deodorized lampante.	Blending oils of different harvest years under a single current year claim.	Direct dilution with seed oils (soy, canola) colored with chlorophyll.
5. Chemical / Measurable	$\text{FFA} \leq 0.3\%$; $\text{PV} \leq 1$ $\text{DAGs} \geq 90\%$; $\text{PPP} \leq 5\%$.	$\text{FFA} \leq 0.8\%$; $\text{PV} \leq 20$; $\text{DAGs} \geq 35\%$;	$\text{FFA} > 0.8\%$; $\text{PV} > 20$; $K_{270} > 0.22$; $\Delta K > 0.01$; wax >150 mg/kg.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
		\text{PPP} \\le 17\%.	
6. Sensory Quality	Vibrant green fruitiness; pronounced herbal bitterness and peppery pungency.	Mild, ripe fruit flavor; faint bitterness; absent or weak peppery sensation.	Sensory defects: fustiness, mustiness, muddy sediment, winy taste, or rancidity.
7. Label-Visible Signals	Specific harvest date; single country/estate origin; dark glass/tin.	Vague multi-country origin; best-by date only; clear glass packaging.	Lacks extra virgin grade; missing harvest date; blended seed oil listings.
8. Label-Invisible Signals	Absolute chemical freshness curve (isomeric DAG conversion over time).	Trace presence of processing solvent residues.	Exact ratio of added refined or foreign vegetable oils.
9. Bari Scoring Signals	Score: +3 for dark glass AND harvest date <12 months; +2 for COOC/PDO.	Score: 0 for standard certified EVOO in dark glass.	Score: -5 for clear glass, missing harvest date, or multi-nation blends.

4. Yogurt

Authentic yogurt is a product of live lactic acid fermentation of milk proteins, driven primarily by *Lactobacillus delbrueckii subsp. bulgaricus* and *Streptococcus*

thermophilus. Exceptional yogurt is crafted from 100% organic, grass-fed whole milk fermented solely with active cultures, containing zero additives, thickeners, stabilizers, starch fillers, or emulsifiers. Straining is performed traditionally to concentrate protein and calcium naturally without starch addition.

Average yogurt relies on conventional, grain-fed pasteurized milk and utilizes added pectins, gums, or gelatin to mimic a thick, strained texture. Poor yogurt is an ultra-processed dairy product containing synthetic stabilizers, colors, and emulsifiers such as polyoxyethylene sorbitan esters, along with artificial sweeteners. It often undergoes post-fermentation pasteurization, which kills the beneficial probiotics to extend shelf life.

Bari can capture organic, grass-fed certifications and scan the ingredient list to immediately penalize stabilizers, gelatins, and artificial sweeteners. It cannot measure the active colony-forming units (CFU/g) at retail, but the engine flags texturizers as a proxy for substandard fermentation times.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	100% organic, grass-fed whole milk with natural fat distribution.	Conventional pasteurized milk from grain-fed dairy herds.	Reconstituted skim milk powder with added water and low-grade whey solids.
2. Production & Processing	Long biological fermentation; natural whey straining; no added heat post-culture.	Rapid industrial inoculation; mechanical thickening using stabilizers.	Post-culture pasteurization to kill live flora; artificial texturizing.
3. Authenticity Indicators	Active live culture claims; zero additives, thickeners, or palm oil.	Standard yogurt labeling containing thickeners like pectin or starch.	"Yogurt-flavored" dairy desserts containing synthetic emulsifiers.
4. Fraud Risks	Adding gelatin or starches to pass off	Reconstituting powdered milk and	Using artificial sweeteners and synthetic colors to

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
	water-diluted milk as thick Greek yogurt.	labeling it as fresh pasteurized dairy.	mask off-flavors in milk.
5. Chemical / Measurable	High active CFU count ($\geq 10^8$ CFU/g); zero gums, starches, or thickeners.	Standard acidity; low active culture count; detectable pectin or gelatin.	High starch content; zero active cultures; presence of polyoxyethylene.
6. Sensory Quality	Velvety, naturally cohesive body; clean, balanced lactic tartness; fresh milk finish.	Firm, gel-like texture; slightly chalky mouthfeel; flat, sweet flavor profile.	Watery or clumpy texture; synthetic fruit flavor; excessively sweet or chemical aftertaste.
7. Label-Visible Signals	Ingredient deck: organic milk and live cultures only; grass-fed certifications.	Pasteurized milk, active cultures, pectin, and cane sugar on the label.	Added "gelatin," "modified food starch," "carrageenan," "aspartame," or colors.
8. Label-Invisible Signals	Exact probiotic strain viability and counts at the point of consumption.	Presence of trace antibiotic residues in the source dairy milk.	Reconstitution ratio of powdered dairy solids.
9. Bari Scoring Signals	Score: +3 for organic milk + active cultures only; +2 for grass-fed seals.	Score: 0 for basic clean-label yogurt with minimal texturizers.	Score: -5 for presence of gelatin, carrageenan, starch, or sweeteners.

5. Cheese

Cheese quality is defined by raw milk biodiversity, enzyme-driven proteolysis, and regional microclimates during aging. Exceptional cheese is produced from raw, grass-fed milk, natural animal rennet, and salt. It is aged traditionally in natural

microclimates to develop complex flavors and amino acid crystallization, and possesses a certified Protected Designation of Origin (PDO/DOP) status.

Average cheese is made from pasteurized, conventional milk using industrial enzyme starters and expedited continuous aging. Poor cheese is an ultra-processed product containing added vegetable oils, sodium phosphate emulsifying salts, and synthetic preservatives. Pre-shredded variations are often heavily coated with wood-derived cellulose powder (as an anti-caking agent) and natamycin to artificially inhibit mold growth.

Bari can capture certified DOP/PDO status, raw milk claims, and the absence of non-cheese ingredients on the deck. It cannot measure non-milk fat substitution or exact proteolysis depth, which requires chromatographic analysis of casein-to-fat ratios, but the engine flags pre-shredded formats as low-quality due to anti-caking additives.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Raw, unpasteurized milk from grass-fed, pasture-raised herds.	Pasteurized conventional milk from grain-fed dairy herds.	Reconstituted dairy powder blended with water, whey solids, and vegetable fats.
2. Production & Processing	Natural animal rennet; traditional hand-crafting; long-duration cave aging.	Standardized industrial cultures; mechanical press curd forming; accelerated aging.	Direct chemical acid coagulation; continuous high-speed extrusion; emulsifier addition.
3. Authenticity Indicators	Certified PDO/DOP or PGI labels with verifiable lot tracing.	Basic varietal name (e.g., Cheddar, Mozzarella) without regional certification.	"Cheese product" or "processed cheese food" labels.
4. Fraud Risks	Substituting raw milk with pasteurized milk; mislabeling country of origin.	Fast-tracking aging claims through the addition of exogenous enzymes.	Blending palm oil or starches into the curd matrix; excessive anti-caking agents.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
5. Chemical / Measurable	High moisture-on-fat-free basis; rich native enzyme profile; zero additive residues.	Moderate salt and moisture levels; zero starch; minimal residual active flora.	Detectable palm fat; high sodium phosphate; high cellulose; natamycin presence.
6. Sensory Quality	Complex, deep savory flavor; tyrosine crystallization; persistent, rich aroma.	Flat, mild flavor; uniform, semi-firm texture; short-lived aroma.	Rubbery, plastic-like texture; chemical or soapy flavor; poor melting performance.
7. Label-Visible Signals	Milk, salt, rennet, cultures only; raw milk claims; DOP/DOP logos.	Pasteurized milk, salt, cheese cultures, enzymes on the ingredient panel.	Added "sodium phosphate," "cellulose," "natamycin," "vegetable oil," or colors.
8. Label-Invisible Signals	Microbial diversity of wild, non-pasteurized bacterial strains in raw milk.	Exact ratio of grass-fed to grain-fed milk lipids.	Presence of trace pesticide bioaccumulation in the cheese fat.
9. Bari Scoring Signals	Score: +3 for active DOP/DOP verification; +2 for "Raw Milk" claims.	Score: 0 for standard clean-label pasteurized cheese.	Score: -5 for presence of emulsifying salts, cellulose, or palm oils.

6. Hummus

Hummus is chemically classified as a ternary oil-in-water colloidal emulsion, where the dispersed fat phase (tahini lipids) is stabilized within an aqueous phase of chickpea starches and proteins using natural emulsifiers. Exceptional hummus

contains organic chickpeas, stone-ground sesame tahini, cold-pressed extra virgin olive oil, fresh lemon juice, and garlic, with zero chemical stabilizers or preservatives.

Average hummus utilizes conventional chickpeas, blends olive oil with cheaper seed oils, and is acidified with citric acid and potassium sorbate (<0.1\%) to achieve a 12 to 15-day shelf life. Poor hummus is diluted with high concentrations of water, uses canola or cottonseed oil instead of olive oil, and contains minimal tahini (<5\%). It is stabilized using up to 1% potassium sorbate (E202) and sodium benzoate (E211). The lack of emulsifier integrity leads to phase separation, and the use of oxidized tahini causes a gritty mouthfeel and metallic, crayon-like off-flavors.

Bari can capture the primary fat source and penalize the presence of potassium sorbate (E202) and sodium benzoate (E211) on the label. It cannot measure the active lecithin concentration or the protein aggregation state, which requires high-performance liquid chromatography (HPLC) and particle-size analysis. The scoring engine mitigates this by flagging hummus that lacks olive oil as a primary fat, as seed oils correlate with higher rates of emulsion collapse and oxidation.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Organic chickpeas; high-grade single-origin sesame tahini; EVOO.	Conventional chickpeas; low-grade tahini; olive oil and canola oil blends.	Low-grade split chickpeas; palm or cottonseed oil; <5\% tahini

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
			content.
2. Production & Processing	High-shear colloidal blending at low temperature ($<8^{\circ}\text{C}$); no post-blend heating.	Standard mechanical grinding; citric acid addition to lower pH; hot-filling.	Water dilution; starch thickening; chemical preservation with artificial stabilizers.
3. Authenticity Index	100% cold-pressed extra virgin olive oil claim; no seed oils or starches.	Olive oil listed behind seed oils; standard food-grade hummus labels.	"Hummus dip" containing

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
Factors			modified starches, gums, and artificial acidulants.
4. Fraud Risks	Substituting expensive sesame tahini with peanut paste or corn starch.	Claiming "Olive Oil Hummus" while using sunflower oil as the primary lipid.	Using heavily oxidized, rancid tahini masked by excess salt and garlic.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
5. Chemical / Measurable	Active lecithin $\geq 0.8\%$ w/w; peroxide value <1.0 ; zero preservatives.	Acidic pH ($4.0\text{--}4.5$); potassium sorbate concentration $<0.1\%$.	Lecithin $<0.3\%$; peroxide value >2.0 ; preservatives up to 1% .
6. Sensory Quality	Silky, non-gritty, creamy texture; sweet nutty sesame flavor; no oil separation.	Slightly firm or paste-like texture; mild sesame flavor; faint acidic sharpness.	Gritty, chalky mouthfeel; phase separation ($\geq 2\text{mm}$ oil ring);

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
			rancid, metallic smell.
7. Label-Visible Signals	EVOO listed as the only oil; organic seals; zero preservatives.	Canola or soybean oil listed alongside olive oil; citric acid; potassium sorbate.	"Vegetable oil," "modified food starch," "potassium sorbate (E202)," "benzoate".
8. Label-Invisible	Particle size of chickpea solids ($<15\mu\text{m}$ preventing grittiness).	exact ratio of sesame tahini used in the pre-blend formulation.	Native protease inhibitor

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
Signals			activity protecting protein solubility.
9. Baring Signals	Score: +3 for EVOO only + zero preservatives; +2 for certified organic.	Score: 0 for standard seed-oil hummus with low preservative levels.	Score: -5 for presence of seed oils, starches, E202, or E211.

7. Tahini

Tahini quality and shelf stability are governed by the oxidative resistance of sesame seed lipids. Sesame oil naturally contains approximately 40% linoleic acid, a polyunsaturated fatty acid highly reactive with oxygen. Exceptional tahini is produced from 100% single-origin sesame seeds (such as the premium Ethiopian Humera cultivar) that are gently toasted at low temperatures and traditionally stone-ground to a particle size of $<15\ \mu\text{m}$. This micro-refinement keeps proteins soluble and yields a sweet, toasted aroma, with a peroxide value of $<1.0\ \text{meq O}_2\text{/kg}$.

Average tahini is made from multi-origin seeds that are roasted rapidly at high temperatures in industrial ovens, causing mild protein denaturation and faster phase separation. Poor tahini is adulterated with cheaper seed oils (such as soybean, rapeseed, or cottonseed oil). It often incorporates over-roasted seeds with burnt chaff, resulting in a bitter, acrid taste, and undergoes advanced oxidative rancidity, producing volatile aldehydes (hexanal, nonanal, and 2,4-decadienal) that smell of old paint or crayons.

Bari can capture single-origin declarations, glass packaging (which provides a 4.2\times lower oxygen transmission rate than plastic), and the presence of 100% sesame seeds. It cannot identify seed oil adulteration markers (like brassicasterol or isoflavones) or quantify hexanal levels without chromatography and mass spectrometry. The engine mitigates this by flagging plastic-packaged tahinis stored in warm retail environments.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	100% single-origin sesame seeds (e.g., Ethiopian Humera cultivar).	Blended multi-origin sesame seeds of variable oil content.	Low-grade, insect-damaged seeds blended with cheaper oilseeds.
2. Production & Processing	Low-temperature toasting; traditional slow stone-grinding below 45°C .	High-speed industrial mechanical milling; standard rapid convective roasting.	Flash-roasting leading to carbonized chaff; high-speed milling causing thermal damage.
3. Authenticity Indicators	100% pure sesame seed claim; packaged in high-barrier glass jars.	Standard pure tahini label; packaged in high-density polyethylene (HDPE).	Blended sesame paste containing added emulsifiers or vegetable oils.
4. Fraud Risks	Diluting tahini with cheap soybean,	Bleaching low-grade dark seeds with chemicals to	Adding peanut paste to simulate rich nuttiness

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
	rapeseed, palm, or cottonseed oils.	pass them off as premium white seeds.	while reducing sesame percentage.
5. Chemical / Measurable	Peroxide value <1.0 meq/kg; zero hexanal; uniform particle size <15\ μm .	Peroxide value <2.0; trace hexanal content; moderate phase separation.	PV >2.0 within 14 days; high hexanal; brassicasterol/sterculic acid presence.
6. Sensory Quality	Creamy, pourable; sweet, nutty, toasted almond aroma; zero acid bitterness.	Viscous, slightly thick; mild roasted flavor; faint, short-lived bitterness.	Waxy, stiff texture; distinct "crayon," paint-like, or "old oil" smell; sharp bitterness.
7. Label-Visible Signals	"100% Sesame Seeds" only; single-origin Humera claims; glass packaging.	"Sesame seeds" on the deck; best-by date only; plastic tub packaging.	Added "vegetable oil," "soy lecithin," "monoglycerides," or preservative codes.
8. Label-Invisible Signals	Absence of soybean isoflavones, rapeseed brassicasterol, or cottonseed malvalic acid.	Exact temperature curve experienced by the sesame seeds during roasting.	Thermodynamic melting profiles (DSC peak shift indicative of palm olein).
9. Bari Scoring Signals	Score: +3 for "Ethiopian Humera" and glass packaging; +2 for clean deck.	Score: 0 for basic clean-label sesame tahini in plastic tubs.	Score: -5 for added oils, emulsifiers, or non-sesame seed ingredients.

8. Honey

Honey is highly susceptible to economically motivated adulteration due to its complex, high-value matrix. Exceptional honey is raw, unpasteurized, and unfiltered, preserving its natural pollen spectrum for botanical and geographical traceabilities. It displays a low hydroxymethylfurfural (HMF) content (<10 mg/kg), verifying that the honey has not been heated to degrade its active enzymes (diastase number >15 , active invertase).

Average honey is pasteurized and finely filtered to prevent crystallization, meeting standard regulatory thresholds (HMF ≤ 40 mg/kg, diastase number ≥ 8). Poor honey is adulterated through the direct addition of C4 sugar syrups (corn, cane) or C3 starch-derived syrups (rice, beet, wheat, cassava, chicory). It may also result from "sugar-feeding fraud," where bees are fed sugar syrup during the active nectar flow. It is often subjected to ultra-filtration and resin-exchange treatments to remove pollen and lighten its color, destroying natural enzymes.

Bari can capture monofloral claims, organic or PDO/PGI certifications, and raw/unfiltered claims while penalizing multi-origin blending statements. It cannot measure HMF, diastase, stable carbon isotope ratios ($\delta^{13}\text{C}$), or identify C3 rice syrup markers (like the 5.39 ppm NMR peak), which requires mass spectrometry and nuclear magnetic resonance. The engine mitigates this by flagging honey blended from non-EU regions as high-risk.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Pure floral nectar gathered naturally by <i>Apis mellifera</i> colonies.	Blended multi-floral nectars; bees supplemented with sugar outside nectar flow.	Exogenous sugar syrups (C3/C4) fed directly to bees during nectar flow.
2. Production & Processing	Cold-extraction below 40°C ; coarse gravity filtration only.	Standard commercial heating and pressure filtration to prevent crystallization.	High-temperature pasteurization; ultra-filtration; ion-exchange resin treatment.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
3. Authenticity Indicators	Monofloral designations; regional PDO/PGI stamps; traceable hive batching.	Pure honey labels with vague country blending declarations.	"Honey blends" containing added corn syrup, rice syrup, or cane sugar.
4. Fraud Risks	Adding industrial starch syrups; ultra-filtration to hide origin.	Mixing cheap imported multi-floral honey into premium monofloral batches.	Artificial coloring (E150d); adding foreign diastase enzymes to fake freshness.
5. Chemical / Measurable	$\text{HMF} < 10 \text{ mg/kg}$; $\text{Diastase} > 15$; $\delta^{13}\text{C}$ protein-honey gap $\leq 1 \text{‰}$	$\text{HMF} \leq 40 \text{ mg/kg}$; $\text{Diastase} \geq 8$; $\delta^{13}\text{C}$ honey-protein gap $< 2 \text{‰}$.	$\text{HMF} > 40 \text{ mg/kg}$; $\text{Diastase} < 8$; isotope gap $\geq 7 \text{‰}$ C4 sugar; NMR 5.39 ppm peak.
6. Sensory Quality	Rich, complex floral aroma reflecting the plant origin; clean, slow-melting sweetness.	Mild, generic sweet taste; absent floral aroma; fast-crystallizing texture.	Flat, sugary, molasses-like sweetness; metallic or fermented off-notes; watery texture.
7. Label-Visible Signals	"Raw, Unfiltered, Unpasteurized" claims; monofloral name; PDO/PGI logos.	"Pure Honey" with multi-national origin statements (e.g., "EU and Non-EU").	"Honey product," added sweeteners, or missing botanical and country origin.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
8. Label-Invisible Signals	Natural pollen fingerprint (melissopalynology origin verification).	Presence of trace neonicotinoid pesticides or tylosin antibiotics.	Dilution with C3 beet or rice syrups below standard isotopic detection limits.
9. Bari Scoring Signals	Score: +3 for "Raw" + monofloral; +2 for PDO/PGI seals.	Score: 0 for basic single-origin pure honey.	Score: -5 for multi-national blends or added syrup ingredients.

9. Coffee (Specialty Arabica)

Specialty coffee is defined by the absence of primary green bean defects and precise roasting thermodynamics. Exceptional specialty coffee consists of 100% Arabica beans scoring ≥ 85 on the Specialty Coffee Association (SCA) 100-point cupping scale, evaluated by a licensed Q Grader. It is roasted in small batches with real-time Rate of Rise (RoR) monitoring, and is dried to a stable moisture range of 10-12% to preserve flavor integrity. Light to medium roasts preserve 60-95% of native antioxidant chlorogenic acids (CGA). The beans are packaged within 24 hours of roasting in high-barrier bags with nitrogen flushing and one-way valves, ensuring freshness for up to 12 months.

Average coffee scores between 80 and 82, lacks farm-level transparency, and is packaged in standard one-way valve bags without nitrogen flushing, reducing its freshness window to 4–8 weeks. Poor coffee uses cheap, ungraded Robusta or defect-heavy commodity beans. These are over-roasted (beyond the second crack, $>435^{\circ}\text{F}$) to mask defects, destroying up to 83% of beneficial CGA and creating a bitter, stomach-irritating beverage.

Bari can capture roast dates, 100% Arabica claims, nitrogen-flushing/one-way valve packaging, and farm-level origin data. It cannot verify the SCA cupping score or post-roast moisture percentage directly, which requires professional sensory panels and colorimeters, but the engine flags coffees that do not print a roast date or use basic packaging.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	100% specialty-grade Arabica; SCA score ≥ 85 ; zero primary defects.	Blend of Arabica beans; SCA score 80–82; minimal physical defects.	Commodity-grade Arabica/Robusta blend; heavy insect, mold, and broken bean defects.
2. Production & Processing	Small-batch roasting; Cropster/Artisan RoR tracking; moisture 10–12%.	Medium-batch industrial roasting; standardized temperature curves; standard moisture.	Continuous automated mega-roasting; high-heat flash roasting; moisture $>13\%$.
3. Authenticity Indicators	Full farm, cooperative, micro-region, altitude ($\geq 1200\text{m}$), and processing method disclosed.	Regional or country-level origin disclosed; standard processing class.	Vague multi-national blends with zero farm-level or altitude transparency.
4. Fraud Risks	Blending cheap Robusta or low-grade Arabica into specialty single-origin lots.	Mislabeling old-crop coffee as current-crop to charge premium prices.	Faking origin (e.g., claiming Blue Mountain or Kona) using cheap commercial beans.
5. Chemical / Measurable	High CGA retention (60–95%); low NMP; moisture 10–12%.	Moderate CGA retention; standard moisture levels.	CGA decreased by 83%; high stomach-irritating N-methylpyridinium (NMP).
6. Sensory Quality	Clean cup; high sweetness; distinct floral, citrus, or chocolate notes; vibrant acidity.	Mild, pleasant coffee flavor; moderate acidity; flat but clean finish.	Strong bitterness; rubber-like, moldy, or musty tastes; dry, astringent mouthfeel.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
7. Label-Visible Signals	Specific roast date (within peak 7-14 day window); single-farm name; nitrogen flush.	"Best By" date with roast date implied; one-way valve bag; country of origin.	No roast date; basic sealed paper bag with no degassing valve or nitrogen flush.
8. Label-Invisible Signals	True SCA cupping score verified by a certified Q Grader panel.	Precise Agtron colorimeter values indicating roast uniformity.	Level of ochratoxin A (mycotoxins) accumulated during storage.
9. Bari Scoring Signals	Score: +3 for Roast Date <30 days; +2 for single-farm tracing.	Score: 0 for generic Arabica with standard one-way valve packaging.	Score: -5 for Robusta blend, missing roast date, or no packaging valve.

10. Chocolate

Chocolate processing is defined by continuous refining and temperature-controlled conching. Exceptional chocolate contains only cacao mass, sugar, and pure cocoa butter rich in symmetric monounsaturated triacylglycerols (SMUT), which provide a clean snap and a smooth melt at body temperature ($32\text{--}35^{\circ}\text{C}$). The solid particles are refined to a size of $<18\ \mu\text{m}$ to ensure a velvety mouthfeel, and are conched extensively to evaporate volatile off-flavors like acetic acid.

Average chocolate uses soy lecithin ($<0.5\%$) to reduce viscosity and save on expensive cocoa butter, refining solids to a standard size of $<25\ \mu\text{m}$. Poor chocolate is compound chocolate where cocoa butter is replaced with palm kernel oil, Cocoa Butter Replacers (CBRs), or Cocoa Butter Substitutes (CBS), leading to poor fat compatibility and a high risk of fat bloom. It relies on high concentrations of polyglycerol polyricinoleate (PGPR, E476) or ammonium phosphatide (E442) emulsifiers to force flow in machinery. It uses heavily alkalized (Dutched) cocoa powders to chemically neutralize off-flavors from low-grade, moldy beans.

Bari can capture emulsifier codes (E476, E442), the presence of vegetable fats, and alkalization declarations on the ingredient deck. It cannot verify the Form V β crystal alignment or the exact particle size distribution, which requires X-ray

diffraction and laser diffraction testing. The engine mitigates this by flagging any chocolate listing PGPR or non-cocoa vegetable lipids.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Single-origin cacao beans; pure, non-deodorized cocoa butter; cane sugar.	Blended cocoa mass; added cocoa butter and soy lecithin (<0.5%).	Low-grade alkalized cocoa powder; palm kernel or hydrogenated vegetable oils.
2. Production & Processing	Extended conching (>24 hours); five-roll refining to particle size <18 μm .	Standard conching (6--9 hours); refining to particle size <25 μm .	High-speed ball milling; zero conching time; artificial chemical emulsification.
3. Authenticity Indicators	"100% Cocoa Butter" claim; non-alkalized; single-origin cooperative tracing.	Standard chocolate labeling; pure cocoa butter and lecithin emulsifiers.	"Chocolate candy" or "chocolate coating" using CBS and CBR fats.
4. Fraud Risks	Substituting cocoa butter with palm-based CBEs without disclosure.	Adding excess lecithin beyond limits to hide a severe reduction in cocoa butter.	Using alkalized cocoa to mask highly acidic, mold-fermented waste beans.
5. Chemical / Measurable	Moisture <0.5%; zero PGPR (E476) or AMP (E442); pure SMUT fat profile.	Moisture <1.0%; soy lecithin <0.5%; zero non-cocoa fats.	High moisture; high PGPR (>0.5%); palm fat presence; elevated alkaline pH.
6. Sensory Quality	Deep glossy finish; hard, sharp snap; fast melting; zero gritty mouthfeel.	Semi-glossy finish; standard snap; slow, creamy melting; slight graininess.	Dull or bloomed surface; soft or bendable snap; waxy, slow-melting mouthfeel.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
7. Label-Visible Signals	Pure cocoa butter; non-alkalized; cocoa percentage $\geq 70\%$; no PGPR.	Cocoa butter, sugar, cacao mass, soy lecithin listed on the ingredient panel.	"Alkalized cocoa," "palm kernel oil," "PGPR (E476)," or "ammonium phosphatide".
8. Label-Invisible Signals	Presence of Form V β crystals indicating precise tempering thermodynamics.	Exact D_{90} particle size distribution of cocoa and sugar solids.	Trace heavy metal accumulation (lead and cadmium) in the cocoa solids.
9. Bari Scoring Signals	Score: +3 for 100% cocoa butter + non-alkalized; +2 for single-origin.	Score: 0 for standard clean-label soy-lecithin chocolate.	Score: -5 for presence of PGPR, CBR/CBS, or palm vegetable fats.

11. Cereal

Breakfast cereal quality is evaluated using carbohydrate-to-fiber ratios. Exceptional cereal is produced from intact, minimally processed whole grains (such as rolled or steel-cut grains) that preserve their natural endosperm, germ, and bran structures. It meets the strict 10:1 total carbohydrate-to-dietary fiber ratio, or the 10:1:1 ratio that restricts added sugars. It is baked gently and packaged using a nitrogen barrier, containing zero synthetic preservatives like BHT or BHA.

Average cereal uses moderately extruded grains, exhibits an intermediate carbohydrate-to-fiber ratio (10:1:2), and contains synthetic antioxidants like BHT, which are typically restricted to the packaging film rather than added directly to the product. Poor cereal is a highly extruded product where twin-screw high-pressure processing compromises the physical structure of the grains. This treatment reduces the short-range order of starch (lowering the Fourier transform infrared spectral absorbance ratio at 1047 cm^{-1} and 1022 cm^{-1}), transforming starches into rapidly digestible starch (RDS) that causes sharp blood glucose spikes. It has a poor carbohydrate-to-fiber ratio ($<10:1$), contains high levels of added sugars, and uses synthetic antioxidants (BHT/BHA) directly in the formulation.

Bari can calculate total carbohydrate-to-fiber-to-sugar ratios from the nutrition facts panel and scan for BHT/BHA on the label. It cannot measure the starch crystallinity ratio or in vivo glycemic index, which requires infrared spectrometry and clinical trials. The engine mitigates this by flagging extruded puffed shapes and rewarding traditional flaked or rolled grains.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Intact whole grains preserving germ, bran, and endosperm; organic sourcing.	Blended whole grain and refined flours; conventional crop sourcing.	Highly refined wheat, corn, or broken rice flours lacking natural fiber.
2. Production & Processing	Gentle rolling, flaking, or low-temperature baking; zero high-pressure extrusion.	Low-shear extrusion; moderate temperature processing; minor starch gelatinization.	High-shear, high-temperature twin-screw extrusion; fully destroyed starch structure.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
3. Authenticity Indicators	Meets 10:1 carbohydrate-to-fiber ratio; zero added sugars or colors.	Meets 10:1:2 carb-to-fiber-to-sugar ratio; standard food-grade claims.	Puffed or extruded shapes claim "whole grain" while containing high sugar.
	4. Fraud Risks	Adding isolated synthetic fibers to refined flour to simulate whole grain content.	Claiming "organic" while sourcing grain from conventional herbicide-treated fields.
5. Chemical / Measurable	Low RDS; high slowly digestible (SDS) and resistant starch (RS); no BHT/BHA.	Moderate RDS; BHT/BHA restricted to the	High RDS; low RS; high glycemic index;

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
		packaging film only.	BHT/BHA added directly to formulation.
	6. Sensory Quality	Naturally crunchy, robust texture; distinct grain-specific flavor; low sweetne ss.	Moderate crispness; quick softening in milk; balanced sweet grain flavor.
7. Label-Visible Signals	Whole grain listed as first ingredient; 10:1 ratio met; zero synthetic additives.	"Whole wheat" first; 10:1:2 ratio; cane sugar on the deck.	Sugar listed near first; <10:1 ratio; "BHT," "BHA," or artificial colors listed.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
8. Label-Invisible Signals	Starch crystallinity ratio (FTIR 1047/1022 cm^{-1} spectral absorbance).	Presence of glyphosate or other agricultural pesticide residues in the grain.	Exact percent age of starch transformed into rapidly digestible fractions.
9. Bari Scoring Signals	Score: +3 for 10:1 or 10:1:1 ratio + whole grain; +2 for organic.	Score: 0 for basic cereal meeting 10:1:2 ratio.	Score: -5 for presence of BHT/BHA, artificial colors, or <10:1 ratio.

12. Pasta

The physics of high-quality dry pasta is governed by extrusion mechanics and starch crystallization. Exceptional pasta is crafted exclusively from high-quality durum wheat semolina (*semola di grano duro*) with a protein content of 11.8% to 12.7% and an optimal gliadin-to-glutenin ratio of 0.8 to 1.0. It is extruded through traditional bronze dies (*trafilati in bronzo*), which create a microscopically rough surface roughness ($R_a \approx 2.8 \mu\text{m}$) that anchors tomato particulates and sauce emulsions. It undergoes slow, low-temperature drying ($\leq 55^\circ\text{C}$) for 48 to 72 hours, preserving the gluten network's tensile strength and keeping the starch in its stable A-

type crystalline form. This results in a wide al dente window and minimal starch leaching during boiling.

Average pasta is bronze-extruded but dried quickly at moderate temperatures (70°C for 8 to 12 hours), which narrows the cooking window. Poor pasta is extruded through Teflon dies, resulting in a glassy, hydrophobic, and sauce-repellent surface ($R_a \approx 0.4 \mu\text{m}$). It is dried rapidly at high temperatures ($85\text{--}105^{\circ}\text{C}$ for only 2 to 4 hours), causing case hardening—where the outer $150 \mu\text{m}$ desiccates quickly—and internal microfractures. When boiled, it leaches excess starch, making the water cloudy and causing the pasta to quickly turn mushy and stick together.

Bari can capture bronze die claims ("trafila in bronzo"), slow-drying claims ("lenta essiccazione"), semola di grano duro, and check if the protein content is $\geq 12\%$. It cannot measure the exact surface roughness (R_a) or starch gelatinization enthalpy directly, which requires laser profilometry and X-ray diffraction. The engine mitigates this by flagging any product that lacks explicit bronze-extruded or slow-dried declarations.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Durum semolina (<i>semola di grano duro</i>); protein 11.8%–12.7%; gliadin-to-glutenin 0.8–1.0.	Standard semolina flour; protein content 11.0%–11.5%.	Low-grade re-milled durum flour (<i>rimacinata</i>); protein $<10.5\%$.
2. Production & Processing	Bronze die extrusion; slow low-temperature drying ($\leq 55^{\circ}\text{C}$), 48–72 hours).	Bronze die extrusion; accelerated oven drying (70°C), 8–12 hours).	Teflon die extrusion; rapid high-temperature drying ($85\text{--}105^{\circ}\text{C}$), 2–4 hours).
3. Authenticity Indicators	"Trafila in bronzo" and "Lenta essiccazione" claims; DOP/IGP Gragnano seals.	Bronze die claims with standard continuous hot-air drying indicators.	No die reference; "extruded" or "shaped" terms masking Teflon use.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
4. Fraud Risks	Labeling Teflon-extruded pasta as bronze-cut; mixing in common soft wheat flour.	Exaggerating drying times on labels (e.g., claiming 48 hours for an 8-hour batch).	"Packed in Italy" claims using cheap imported non-durum wheat flour.
5. Chemical / Measurable	Surface roughness $R_a \approx 2.8 \mu\text{m}$; high starch gelatinization enthalpy.	Surface roughness $R_a \approx 1.5 \mu\text{m}$; moderate starch leaching.	Roughness $R_a \approx 0.4 \mu\text{m}$; gelatinization enthalpy decreased by 42%; high starch leaching.
6. Sensory Quality	Rough matte texture; perfect al dente firmness; wide cooking window; holds sauce.	Slightly smooth texture; moderate chew; narrow cooking window; average sauce grip.	Slippery, glassy texture; quick transition from undercooked to mushy; sauce pools at bottom.
7. Label-Visible Signals	"Trafila in bronzo"; "lenta essiccazione"; protein $\geq 12\%$; PGI/IGP logos.	"Bronze extruded"; best-by date only; protein 11.0%–11.5% on nutrition panel.	"Semolina flour" or "wheat flour"; protein $< 11\%$; no extrusion or drying claims.
8. Label-Invisible Signals	Exact microscopic hydration rate of the pasta surface during boiling.	Purity of the durum wheat cultivar (e.g., Svevo vs. blended cultivars).	Microfractures in the starch-protein matrix caused by high-heat case hardening.
9. Bari Scoring Signals	Score: +3 for "trafila in bronzo" + "lenta	Score: 0 for basic bronze-extruded	Score: -5 for protein $< 11\%$ or missing

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
	essiccazione" + protein \ge 12\%.	pasta without drying claims.	extrusion declarations.

13. Canned Fish

The quality of canned fish is a matter of scombroid safety, heavy metal bioaccumulation, and oil medium purity. Exceptional canned fish consists of hand-packed premium cuts (such as *Ventresca* or belly) preserved in high-quality extra virgin olive oil. It has extremely low histamine levels (<2 ppm), indicating the fish was frozen immediately upon harvest to prevent bacterial spoilage, and is packed in BPA-free lined cans to prevent heavy metal leaching. It carries Marine Stewardship Council (MSC) or Dolphin Safe certifications.

Average canned fish consists of solid or chunk cuts packed in standard refined olive oil or water, showing histamine levels <10 ppm. Poor canned fish consists of shredded or flaked fish packed in brine or cheap seed oils. It displays signs of pre-processing spoilage with histamine levels exceeding 50 mg/kg, posing a risk of scombroid poisoning. It contains elevated levels of lead and tin due to poor can lining and long storage times, and has a mushy, waterlogged texture with strong, unpleasant fishy off-odors.

Bari can capture the specific cut type, the packing medium (EVOO vs. seed oil), and the presence of sustainability seals (MSC, Dolphin Safe). It cannot verify the exact histamine concentration or trace heavy metal burden (lead, tin, cadmium, or mercury) directly, which requires high-performance liquid chromatography (HPLC) and Inductively Coupled Plasma Mass Spectrometry (ICP-MS). The engine mitigates this by flagging shredded or flaked formats packed in water or seed oils.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
1. Raw Materials	Hand-packed premium cuts (e.g., <i>Ventresca</i> belly fillets); sustainably harvested fish.	Solid chunks or whole loins; standard commercial wild-caught fish.	Machine-packed shredded flakes, scraps, or processing waste.

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
2. Production & Processing	Immediate freezing at sea; manual dressing and cleaning; hand-packing.	Quick blast freezing; automated mechanical skinning and loin cutting.	Prolonged storage at warm temperatures prior to canning; scombroid breakdown.
3. Authenticity Indicators	100% Extra Virgin Olive Oil pack; MSC or Dolphin Safe certified logos.	Pure olive oil or water pack; standard food-safe canned fish labeling.	Preserved in cheap soybean, sunflower, or generic vegetable seed oils.
4. Fraud Risks	Substituting expensive species (e.g., Bluefin) with cheaper species (e.g., Skipjack).	Claiming "Olive Oil Pack" when seed oils make up the bulk of the medium.	Canning spoiled, high-histamine fish masked by excessive salt and sterilization.
5. Chemical / Measurable	$\text{Histamine} < 2$ ppm; cadmium/lead below detection limits; BPA-free lining.	$\text{Histamine} < 10$ ppm; standard low heavy metal concentrations; tin $\leq 10 \text{ } \mu\text{g/g}$.	$\text{Histamine} \geq 50$ mg/kg; high tin, lead, or mercury leaching; salt $> 2.0\%$.
6. Sensory Quality	Firm, tender fillets; clean, delicate fish flavor; rich, aromatic olive oil finish.	Semi-firm chunks; neutral flavor; mild, standard fishy aroma.	Mushy, waterlogged texture; metallic aftertaste; strong, unpleasant fishy off-odors.
7. Label-Visible Signals	"Ventresca" or "belly fillets"; EVOO listed as the only oil; MSC logo; BPA-Free.	"Solid" or "chunk" light tuna; olive oil or spring water; best-by date.	"Flakes," "shredded," or "grated"; "vegetable oil" or soybean oil; high sodium.
8. Label-Invisible Signals	Exact histamine ppm level indicating pre-	Trace concentrations of bioaccumulated	Presence of polycyclic aromatic

Parameter / Dimension	Exceptional Standard	Average Standard	Poor Standard
	processing temperature control.	methylmercury or microplastics.	hydrocarbons (PAHs) if wood-smoked.
9. Bari Scoring Signals	Score: +3 for "Ventresca" AND EVOO pack; +2 for MSC/Dolphin Safe.	Score: 0 for solid chunks in standard olive oil or water.	Score: -5 for shredded/flaked fish, seed oils, or salt >1.5\%.

The Bari Scoring Engine Framework Implementation

To operationalize the intelligence gathered across these thirteen categories, the Metropolitan City of Bari can deploy an automated, OCR-enabled parsing algorithm. This digital engine scans retail packaging barcodes and ingredient decks, matching them against regional quality standards, processing parameters, and nutritional profiles.

Analytical Capabilities: Realistically Capturable vs. Invisible Signals

The table below contrasts the realistically capturable signals that the digital engine can evaluate against the invisible parameters that require laboratory testing, showing how the engine uses risk-mitigation proxies to account for these invisible factors.

Food Category	Realistically Capturable Signals (Via Digital Engine)	Invisible Signals (Requires Laboratory Testing)	Bari Digital Mitigation Proxy (Risk Flagging)
Bread (Sourdough)	• Presence of commercial yeast • Souring additives (citric acid) • Preservative list	• Live sourdough microflora CFU • True lactic-to-acetic acid ratio	Flag products that do not explicitly claim "100% natural fermentation" or "long cold proof".
Butter	• Added emulsifiers/stabilizers • Certified grass-fed/organic claims	• Fatty acid profile (CLA, Omega ratio) • Palm-derived fat blending	Apply a risk flag to conventional, non-organic butter lacking grass-fed certifications.

Food Category	Realistically Capturable Signals (Via Digital Engine)	Invisible Signals (Requires Laboratory Testing)	Bari Digital Mitigation Proxy (Risk Flagging)
Olive Oil	• Dark glass packaging • Harvest-to-sale age • DOP/PGI/COOC certifications	• FFA, PV, UV absorbency (K₂₇₀) • DAGs & PPP values	Heavily penalize clear glass packaging and products lacking a specific harvest date.
Yogurt	• Stabilizer/thickener additives • Total sugars per portion	• Probiotic cell viability (CFU/g) • Synthetic growth hormones	Penalize products listing gelatin, pectin, or starches, as they suggest accelerated fermentation.
Cheese	• Anti-caking agents (cellulose) • DOP/IGP certification status	• Casein integrity and proteolysis • Pasteurized vs. raw microflora	Check DOP registries; flag processed variations and products with non-milk fats.
Hummus	• Oil type (EVOO vs. seed oils) • Preservatives (E202/E211)	• Lecithin concentration (w/w%) • Pre-blend lipid oxidation levels	Flag the use of cheap seed oils, which correlate with high oxidation risk and emulsion failure.
Tahini	• 100% sesame seed purity • Glass vs. plastic packaging	• Seed oil adulteration markers • Peroxide and hexanal levels	Flag plastic packaging stored in warm retail environments to mitigate rancidity risks.

Food Category	Realistically Capturable Signals (Via Digital Engine)	Invisible Signals (Requires Laboratory Testing)	Bari Digital Mitigation Proxy (Risk Flagging)
Honey	• Monofloral/PGI/PDO claims • Non-EU multi-origin blending	• Stable C3/C4 isotope ratio • 5.39 ppm NMR rice syrup peak	Apply high-risk flags to multi-origin, non-certified blends from high-fraud import hubs.
Coffee	• Roast date freshness window • Traceability depth	• SCA cupping score • Post-roast moisture %	Penalize products that do not disclose their harvest year, processing method, or SCA score.
Chocolate	• Added PGPR or CBS fats • Alkalized ("Dutched") claims	• Exact particle size (D_{90}) • Crystal polymorph stability	Flag products listing PGPR (E476) or CBS fats, as they indicate poor refining and low fat quality.
Cereal	• Carb-to-fiber ratio (10:1) • Added synthetic BHT/BHA	• Starch crystallinity ratio • Glycemic index response in vivo	Calculate carbohydrate quality metrics; penalize highly extruded, low-fiber shapes.
Pasta	• "Trafila in bronzo" claims • "Lenta essiccazione" claims	• Physical surface roughness (Ra) • Starch gelatinization damage	Penalize products that do not explicitly claim both "bronze extrusion" and "slow drying".
Canned Fish	• Fillet type vs. shredded flakes • Packing medium (EVOO vs. water)	• Histamine ppm concentration • ICP-MS heavy metal leaching	Flag shredded or flaked fish packed in water or seed oils due to a higher risk of high histamine.

Operational Recommendations for Bari

To deploy and maintain this Category Excellence Intelligence Framework, the Metropolitan City of Bari should focus on three strategic areas :

- 1. Integrate Regional Quality Registries:** Bari can directly link its digital scoring engine with the regional "Prodotti di Qualità di Puglia" voluntary quality scheme and European Union PDO/PGI databases. This integration enables the scoring engine to instantly verify geographic authenticity and trace products back to their certified origin, helping to mitigate fraud risks in high-stakes categories like olive oil, cheese, and honey.
- 2. Implement Automated Nutrition and Processing Auditing:** By analyzing nutrition facts panels, the framework can automatically calculate carbohydrate quality ratios (10:1 and 10:1:1) for cereals and breads. This allows the system to identify highly processed, high-glycemic extruded products and flag them accordingly. Simultaneously, the system can scan ingredient decks to penalize the use of synthetic antioxidants (BHT/BHA), thickeners, and emulsifiers (such as PGPR and carrageenan) across all dairy and chocolate categories.
- 3. Deploy Label-Visible Proxies for Invisible Food Risks:** While advanced laboratory metrics—such as olive oil DAGs, honey carbon isotope ratios, and canned fish histamine levels—cannot be directly captured from retail labels, Bari can implement digital mitigation proxies. The scoring engine can flag products packaged in high-risk materials (like clear glass for olive oil or plastic for tahini) or those sourced from regions with a history of supply chain fraud. These proxies protect consumers from oxidation, degradation, and adulteration before laboratory testing is required.

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